# Mriganko Shekhar Dutta

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## Research Interests

I am broadly interested in Computer Architecture. This includes advanced topics such as processor frontend design and CPU pipeline-memory hierarchy interaction.

## Education

**Indian Institute of Technology Bombay, Mumbai** *(Jul 2024 - present)*
*Master of Science by Research in Computer Science and Engineering* GPA: 8.54/10

**Techno India University, Kolkata** *(Oct 2020 - Jul 2024)*
*Bachelor of Technology in Computer Science and Engineering* GPA: 9.28/10

## Research Work

**Designing a High-Performance Cache Management Policy for Datacenter Processors**
*(Prof. Biswabandan Panda)* *(Feb 2025 - present)*

- Studied the effects of cache misses on the frontend and the backend of the processor pipeline
- Analyzed various cache management techniques, aimed at reducing frontend or backend stalls for processors
- Observed the increase in backend pressure with the instruction-aware L2 replacement policies
- Developed a **novel** L2 cache replacement policy, SWING, that **addresses both frontend and backend stalls** by identifying and prioritizing *critical(costly)* cache misses at L2, based on the program execution behavior
- Implemented and evaluated the SWING replacement policy on gem5 simulator, resulting in a geomean speedup of 14.90% with a storage overhead of 2.25KB, while the state-of-the-art replacement policy ICARUS (ASPLOS '26) resulted in 12.98%

**Understanding and Analyzing Transient Execution Attacks**
*(Prof. Biswabandan Panda)* *(Aug 2024 - Jan 2025)*

- Studied the working of transient execution attacks and microarchitectural side and covert channel attacks
- Analyzed the implementation of GhostMinion (MICRO '21) mitigation for Spectre attack (S&P '18)
- Extended the idea of GhostMinion cache to the L1 instruction cache to prevent transient execution attacks through the frontend of the processor pipeline

## Course Projects

**Microarchitecture Aware Performance Optimization**
*(Prof. Biswabandan Panda | CS683:Advanced Computer Architecture)*

- Optimized 2D-convolution operation. Achieved 4-5x speedup using **blocked matrix multiplication, SIMD software prefetching on a real system**
- Implemented **IP-stride** and **complex-stride-prefetcher** in the Champsim simulator

**Rowhammer Attack on DNN Models**
*(Prof. Biswabandan Panda | CS773: Computer Architecture for Performance and Security)*

- Analyzed PyTorch library code to find out vulnerable attack surfaces
- Profiled different image classification models like ResNet, DenseNet to find out vulnerable weight locations
- Performed rowhammer attack on a victim process loading the DNN model. Evaluation of ResNet 152 on ImageNet resulted in accuracy drop from 70% to less than 1%.

**Distributed Key-Value Store**
*(Prof. Purushottam Kulkarni | CS733: Design and Engineering of Computing Systems)*
- Designed and implemented a distirbuted key-value store with multiple connection support and a caching mechanism at each node.

## Experience

**IITB Trust Lab** **Mumbai**
*System Administrator* *(Sep 2025 - present)*
Maintaining and managing servers and virtual environments used by 35+ researchers' infrastructure.

**Chegg India** **Remote**
*Subject Matter Expert* *(May 2022 - Oct 2022)*
Solved academic doubts and cleared concepts in computer science and engineering for students around the globe registered on the Chegg platform.

## Technical Skills

**Languages**: C++, SQL, Bash, Python, Assembly(8085)
**Simulators**: Gem5, Champsim, PCACTI, Vensim

## Teaching

Served as a TA for CS683: Advanced Computer Architecture *(Aug 2025 - Dec 2025)*

## Courses Undertaken

- CS794: Systems for Machine Learning
- CS752: System Dynamics
- CS773: Computer Architecture for Performance and Security
- CS683: Advanced Computer Architecture
- CS733: Design and Engineering of Computing Systems
- CS684: Embedded Systems
- CS699: Software Engineering Lab

## Awards and Test Scores

- Silver medalist in B. Tech, CSE department, Techno India University, Kolkata
- AIR 645 out of 1,23,967 candidates in GATE CS 2024